

RODRIGO GOMES DE ARAÚJO

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PROFILE

MSc Computer Science student graduating in **July 2027**, interested in **financial systems** and the reliable and secure software that powers them. Experience in **Rust, Python and PostgreSQL**, with projects spanning a decentralized transaction ledger, an analytical data platform and machine-learning prediction. Thesis work on protocols that are **correct and secure by construction**, useful for financial transaction workflows. Rigorous, reliable and comfortable building systems where correctness matters. Based in Porto and available to work in Lisbon or Remote.

EDUCATION

Master's <i>Computer Science</i>	Sep 2025 – Present (Jul 2027) <i>University of Porto (FCUP)</i>
Bachelor's <i>Computer Science and Engineering</i>	Sep 2022 – Jul 2025 <i>University of Porto (FEUP)</i>

RESEARCH

CaST: Cryptographically-aware Session Types <i>Master's Thesis</i>	Jul 2026 – Present (Jul 2027) <i>FreeST, Haskell, Type Systems, Applied Cryptography</i>
- Researching how to make communication protocols correct and secure by construction, so invalid cryptographic operations are rejected before deployment. - Developing formal models and compiler support for tracking authorization and cryptographic evidence across nested protocols, with applications to financial transaction and smart-contract workflows.	

PROJECTS

PoW Blockchain for Decentralized Auctions <i>Data Systems Security · 19/20</i>	<i>Rust, Tokio, LibP2P, Docker, Blake2b, Ed25519</i>
- Built a decentralized public ledger for auctions, where signed transactions are validated by network participants and recorded in an immutable chain. - Implemented proof-of-work consensus, transaction validation, Merkle-tree block commitments and Kademlia peer discovery. - Automated multi-node Docker simulations with honest and malicious participants, developing skills in distributed trust, fault resilience and adversarial testing.	

Portuguese Elections Analytics Platform <i>Advanced Topics in Databases · 18/20</i>	<i>PostgreSQL, PostGIS, Python, SQL, OCaml</i>
- Built an end-to-end analytical platform for Portuguese election results, from Python ETL of official Excel data into PostgreSQL to an interactive visual application. - Designed a star-schema data warehouse for votes, turnout and seats, with spatial data and indexes supporting analysis from national to parish level. - Implemented analytical SQL for trend, vote-swing, turnout and party-performance analysis, using roll-ups, CUBE, window functions and geospatial queries.	

UFC Fight Outcome Prediction System <i>Artificial Intelligence · 20/20</i>	<i>Python, pandas, NumPy, scikit-learn, Streamlit</i>
- Built an end-to-end machine-learning system to predict fight outcomes from historical data, engineering 30+ predictive features from performance, experience and physical statistics. - Trained and evaluated six models. Logistic Regression, Random Forest, Gradient Boosting, SVM, Neural Network and a soft-voting ensemble, using five-fold cross-validation and explicit overfitting checks. - Built an interactive Streamlit application with probability estimates, model comparisons, feature analysis and visual reports.	

TECHNICAL SKILLS

Programming & Data: Rust, Python, SQL, PostgreSQL, C/C++, Haskell, OCaml, Java
Data & Systems: Tokio, LibP2P, Tonic, pandas, NumPy, scikit-learn, PostGIS, Git, UNIX, Docker, Terraform, OpenMP
Spoken Languages: Portuguese (Native), English (B2 First, Cambridge), Spanish (Proficient)

AWARDS

2nd Place in IEEE RetroJam 2025 IEEE UP Student Branch	Oct 2025
Built a game from scratch in 48 hours using Rust and Raylib.	
Academic Merit Awards	2020, 2021, 2022
Participation in RoboCup 2016 RoboCup Federation	Jun 2016
1st Place in the National Robotics Championship Instituto Politécnico de Bragança	May 2016